preserves $\hat{\rho}(\xi) = \exp(-\beta\mu\xi^2/2)$. However, this is obvious from our earlier investigations of Ornstein-Uhlenbeck equations.

In order to show that this thermostat is ergodic, we need to demonstrate that the assumptions presented in Sect. 6.4.4 are valid. Assumption 1(ii) requires us to verify that a Hörmander condition (see Definition 6.1) holds for the Nosé-Hoover-Langevin system. Because of the complexity of the high order commutators, we will work here with the assumption of a quadratic potential in the physical model. In some sense this is the most difficult case for a thermostat, but paradoxically, the assumption facilitates the mathematical analysis. We will assume throughout the following that $\mathrm{N_d} = \mathrm{N_c}$, i.e. that there are no constraints in the system. Our potential will therefore be assumed to be

$$U(\boldsymbol{q}) = \frac{1}{2}\boldsymbol{q}^T\boldsymbol{A}\boldsymbol{q},$$

for $\boldsymbol{A}$ a symmetric, positive definite $\mathrm{N_c} \times \mathrm{N_c}$ matrix, with

$$\nabla U(\boldsymbol{q}) = \boldsymbol{A}\boldsymbol{q}.$$

We will further assume $\boldsymbol{M} = \boldsymbol{I}$. If this is not the case, then one may introduce the rescaling $\tilde{\boldsymbol{q}} = \boldsymbol{M}^{-1/2}\boldsymbol{q}, \tilde{\boldsymbol{p}} = \boldsymbol{M}^{1/2}\boldsymbol{p}$ in order to rewrite the system in a form that does not involve $\boldsymbol{M}$. Thus the system under study is

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{p}\mathrm{d}t, \tag{8.30}$$

$$\mathrm{d}\boldsymbol{p} = -\boldsymbol{A}\boldsymbol{q}\mathrm{d}t - \xi\boldsymbol{p}\mathrm{d}t, \tag{8.31}$$

$$\mathrm{d}\xi = \mu^{-1}[\|\boldsymbol{p}\|^2 - \mathrm{N_d}\mathrm{k_B}\mathrm{T}]\mathrm{d}t - \gamma\xi\mathrm{d}t + \sigma\mathrm{d}\mathrm{W}. \tag{8.32}$$

Let $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{\mathrm{N_c}}$ be the eigenvectors of $\boldsymbol{A}$ with $\lambda_1, \ldots, \lambda_{\mathrm{N_c}}$ the corresponding eigenvalues. Observe that if $\boldsymbol{q}_0 \cdot \boldsymbol{u}_k = \boldsymbol{p}_0 \cdot \boldsymbol{u}_k = 0$, then, using (8.30)–(8.31), we have at this point

$$\boldsymbol{u}_k \cdot \boldsymbol{p} = 0, \qquad \boldsymbol{u}_k \cdot [-\boldsymbol{A}\boldsymbol{q} - \xi\boldsymbol{p}] = -\lambda_k \boldsymbol{u}_k \cdot \boldsymbol{q} = 0,$$

which implies that $V_k = \{(\boldsymbol{q}, \boldsymbol{p}) | \boldsymbol{u}_k \cdot \boldsymbol{q} = 0, \boldsymbol{u}_k \cdot \boldsymbol{p} = 0\}$ is an invariant submanifold. Therefore we must require that $\boldsymbol{u}_k \cdot \boldsymbol{q}$ and $\boldsymbol{u}_k \cdot \boldsymbol{p}$ do not simultaneously vanish, that is, $(\boldsymbol{q}, \boldsymbol{p}) \in \mathcal{D} = \mathbb{R}^{2\mathrm{N_c}} \setminus \bigcup_{k=1}^{\mathrm{N_c}} V_k$.

Theorem 8.1 *If A is symmetric positive definite and has distinct eigenvalues, then (8.30)–(8.32) satisfies the Hörmander condition on $\mathcal{D} \times \mathbb{R}$.*

Proof Observe that Nosé-Hoover-Langevin dynamics is a special case of the system (8.25)–(8.26) Applying the proposition in our case, the problem is to show that the Lie algebra generated by

$$\boldsymbol{F} = \begin{bmatrix} \boldsymbol{p} \\ -\boldsymbol{A}\boldsymbol{q} \end{bmatrix}, \qquad \boldsymbol{G} = \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{p} \end{bmatrix},$$

spans $\mathbb{R}^{2\mathrm{N_c}}$. We shall write $S\{\boldsymbol{F}, \boldsymbol{G}\}$ for the Lie algebra generated by $\boldsymbol{F}$ and $\boldsymbol{G}$.

Next we note:

Lemma 8.1 *For* $k = 1, 2, \ldots,$

$$\boldsymbol{C}_k = \begin{bmatrix} \boldsymbol{A}^{k-1}\boldsymbol{p} \\ \boldsymbol{A}^k\boldsymbol{q} \end{bmatrix} \in S\{\boldsymbol{F}, \boldsymbol{G}\},$$

and

$$\boldsymbol{D}_k = \begin{bmatrix} \boldsymbol{A}^k\boldsymbol{q} \\ -\boldsymbol{A}^k\boldsymbol{p} \end{bmatrix} \in S\{\boldsymbol{F}, \boldsymbol{G}\}.$$

The proof of this is straightforward by induction on k. (See exercises.)
The conclusion of the proof of the theorem is based on the following proposition:

Proposition 8.3 *Assume* $\boldsymbol{A}$ *to be a symmetric positive definite matrix with distinct eigenvalues, and let* $\boldsymbol{C}_k$, $\boldsymbol{D}_k$ *be defined as in Lemma 8.1, then the* $2\mathrm{N_c}$ *vectors* $\boldsymbol{C}_1, \boldsymbol{D}_1, \boldsymbol{C}_2, \boldsymbol{D}_2, \ldots, \boldsymbol{C}_{\mathrm{N_c}}, \boldsymbol{D}_{\mathrm{N_c}}$ *form a linearly independent set on* $\mathcal{D}$.

Proof $\boldsymbol{A}$, being symmetric positive definite is unitarily similar to a diagonal matrix $\boldsymbol{\Lambda}$, so $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^T$. Define $\hat{\boldsymbol{p}} = \boldsymbol{U}^T\boldsymbol{p}$, $\hat{\boldsymbol{q}} = \boldsymbol{U}^T\boldsymbol{q}$. The domain $\mathcal{D}$ is defined by the condition that $\hat{p}_i$ and $\hat{q}_i$ are not simultaneously zero.

With the change of variables, the problem is to show that $\hat{\boldsymbol{C}}_1, \hat{\boldsymbol{D}}_1, \ldots, \hat{\boldsymbol{C}}_{\mathrm{N_c}}, \hat{\boldsymbol{D}}_{\mathrm{N_c}}$ are linearly independent, where

$$\hat{\boldsymbol{C}}_k = \begin{bmatrix} \boldsymbol{\Lambda}^{k-1}\hat{\boldsymbol{p}} \\ \boldsymbol{\Lambda}^k\hat{\boldsymbol{q}} \end{bmatrix}, \qquad \hat{\boldsymbol{D}}_k = \begin{bmatrix} \boldsymbol{\Lambda}^k\hat{\boldsymbol{q}} \\ -\boldsymbol{\Lambda}^k\hat{\boldsymbol{p}} \end{bmatrix}.$$

Suppose the vectors are linearly dependent, then there exist coefficients $\alpha_0, \alpha_1, \ldots, \alpha_{\mathrm{N_c}-1}$ and $\beta_0, \beta_1, \ldots, \beta_{\mathrm{N_c}-1}$ not all zero such that:

$$\sum_{j=1}^{\mathrm{N_c}} \alpha_j \hat{\boldsymbol{C}}_j + \sum_{j=1}^{\mathrm{N_c}} \beta_j \hat{\boldsymbol{D}}_j = 0.$$

Define polynomials $\pi(x) = \sum_{j=0}^{\mathrm{N_c}-1} \alpha_j x^j$, $\sigma(x) = \sum_{j=0}^{\mathrm{N_c}-1} \beta_j x^j$, then, for each j, we have

$$\pi(\lambda_j)\hat{q}_j + \lambda_j\sigma(\lambda_j)\hat{p}_j = 0, \qquad \lambda_j\pi(\lambda_j)\hat{p}_j - \lambda_j\sigma(\lambda_j)\hat{q}_j = 0,$$

multiplying the first equation by $\hat{q}_j$ and the second by $\hat{p}_j$ and adding, we obtain

$$[\hat{q}_j^2 + \lambda_j\hat{p}_j^2]\pi(\lambda_j) = 0.$$

The first factor is nonzero by the condition of being in $\mathcal{D}$ and the positive definiteness of $\boldsymbol{A}$, thus we must have $\pi(\lambda_j) = 0$ for all $\mathrm{N_c}$ distinct λ_j. But as π has degree $\mathrm{N_c} - 1$, this is a contradiction. Thus the collection must be linearly independent.

Demonstrating the Hörmander condition is sufficient for Assumption 1(ii) in Sect. 6.4.4. Assumption 1(i) can be verified by proving that solutions are able to access an open set around any point $\boldsymbol{y} \in \mathcal{D} \times \mathbb{R}$. We omit the proof of Assumption 1(i) for Nosé-Hoover-Langevin dynamics as it is follows the same structure as the proof of Lemma 6.1 for Langevin dynamics. The details of the proof can be found in [283], with examples for more general SDEs given in [256, 257].

The final requirement for geometric ergodicity (specified in Assumption 2 in Sect. 6.4.4) requires a function $\phi : \mathcal{D} \times \mathbb{R} \to (0, \infty)$ such that

$$\mathcal{L}\phi \leq -\alpha\phi + \delta,$$

for positive constants α and δ and where $\mathcal{L}$ is the generator of the stochastic process. Such a function is called a Lyapunov function.

The generator for Nosé-Hoover-Langevin dynamics (8.27)–(8.29), acting on a suitable test function $f(\boldsymbol{q}, \boldsymbol{p}, \xi)$ is

$$\mathcal{L}_{\mathrm{NHL}} f = \boldsymbol{M}^{-1}\boldsymbol{p} \cdot \nabla_q f - \nabla U(\boldsymbol{q}) \cdot \nabla_p f - \xi \boldsymbol{p} \cdot \nabla_p + \mu^{-1}\left[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d}\mathrm{k_B}\mathrm{T}\right]\frac{\partial f}{\partial \xi}$$
$$- \gamma\xi\frac{\partial f}{\partial \xi} + \mathrm{k_B}\mathrm{T}\gamma\mu^{-1}\frac{\partial^2 f}{\partial \xi^2}.$$

If we consider $\boldsymbol{M} = \boldsymbol{I}$ for simplicity, a Lyapunov function is given in [283] as

$$\phi(\boldsymbol{q}, \boldsymbol{p}, \xi) = \frac{1}{2}\|\boldsymbol{p}\|^2 + U(\boldsymbol{q}) + \frac{\mu}{2}(\xi - a)^2,$$

where a is a constant. As our potential is smooth, it is reasonable to make the assumption that

$$1 < U_{\min} \leq U(\boldsymbol{q}) \leq U_{\max} < \infty, \quad \forall \boldsymbol{q} \in \mathcal{Q},$$